ARC-SPEC-2026-EDU-03: Industrial Accreditation, Institutional Integration & Faculty Onboarding

A Protocol for Asynchronous Institutional Integration and Industrial Certification of DAE Curricula

Jesse Tuohy (Principal Invariant Architect)
Arcstone Adaptive Science Systems

Status: Archive-Ready / Path A Academic Extension
Governing Invariants:
$C_{\text{ops}} = 0 \quad | \quad \tau_{\text{override}} \leq 11.99\text{ms} \quad | \quad \text{Administrative Drag } (A_{\text{drag}}) = 0$
Target Baseline: Phase 7 Realization (`v1.3.0-exec`)

---

1 Executive Summary & Institutional Strategy

While ARC-SPEC-2026-EDU-01 establishes the mathematical theory and ARC-SPEC-2026-EDU-02 provides the automated verification harness, this specification solves the institutional adoption bottleneck.

Traditional academic curriculum integration is notorious for severe administrative inertia: multi-year faculty review boards, departmental politics, accreditation audits, and manual syllabus redesigns. For an authorization-first, sub-12ms engineering framework to proliferate, institutional adoption must occur without introducing synchronous drag on either the academic institution or the core architecture:

$$\text{ADOPTION INVARIANT: } A_{\text{drag}} = 0 \quad \Longrightarrow \quad \frac{\partial C_{\text{ops}}}{\partial \text{University}_{\text{count}}} = 0$$

ARC-SPEC-2026-EDU-03 formalizes the Asynchronous Institutional Integration Protocol (AIIP). It provides a turn-key mechanism for university computer science and electrical engineering departments to map DAE modules directly into existing ABET/IEEE-CS accredited course structures, issue verified industrial micro-credentials, and onboard faculty asynchronously via automated, self-evaluating CI pipelines.

Adoption Protocol Comparison

TRADITIONAL CURRICULUM ADOPTION (High Drag / Multi-Year Lag):

[Faculty Proposal] $\longrightarrow$ [Board Reviews] $\longrightarrow$ [Accreditation Audits] $\longrightarrow$ [Manual Syllabus Rewrite]

DAE ASYNCHRONOUS ADOPTION PROTOCOL (AIIP / Zero Drag):

[Module Overlay] $\longrightarrow$ [Fork Path A Core] $\longrightarrow$[Run substrate-ci.yml] $\longrightarrow$ [Auto ABET Map]
$\Downarrow$
$\longrightarrow$ [Auto-Issued Course Badges]

2 ABET & IEEE-CS Curriculum Mapping Matrix

To eliminate the need for university departments to rewrite existing accreditation documentation, DAE modules map directly into standard ABET Student Outcomes (1 through 7) for Computer Science, Computer Engineering, and Software Engineering.

Table 1: ABET Student Outcomes Mapping Matrix

DAE Module	ABET Criterion	Primary Artifact Produced
Module I: Ingress Admissibility	SO-2 (Engineering Design)	Signed XDP Kernel Module (`.o`)
Module II: Dynamic State Space	SO-1 (Complex Engineering)	5D Vector Drift Monitor
Module III: Hardware Interlocks	SO-6 (Experimentation)	Sub-12ms HIL Interlock Test Log
Module IV: Bounded Cybernetics	SO-7 (Acquire Knowledge)	Kingman Queue Clamps ($\rho \leq 0.80$)

2.1 Turn-Key ABET Syllabus Integration Clause

Departmental chairs can insert the following standardized clause directly into existing Senior/Graduate Systems Elective syllabi (e.g., CS-4000/5000 series Advanced Operating Systems or Real-Time Embedded Systems):

> *"This course incorporates the Deterministic Admissibility Engineering (DAE) framework (ARC-SPEC-2026-EDU-01). Students will complete four automated laboratory verification modules evaluating sub-12ms execution ceilings, zero-heap memory bounds, and 5D state-vector stability using bare-metal Rust (`#![no_std]`) and eBPF/XDP kernel structures."*

3 Asynchronous Faculty Onboarding

To ensure university professors can adopt the curriculum without attending manual training seminars or revising slides, the DAE courseware is packaged as a Plug-and-Play Educational Container (PPEC) inside the open-core repository (`/docs/academic/courseware`).

```
# Faculty Verification Execution Command
git clone https://github.com/trencinodin-stack/arcstone-continuity-core
    .git
cd arcstone-continuity-core/docs/academic/harness
cargo run --bin faculty_bootstrap_verify -- --spec ARC-SPEC-2026-EDU-03
```

Listing 1: Faculty Bootstrap Execution Command

The runner verifies that local hardware microarchitectures support precise cycle counters (`RDTSC/TSC-INVARIANT`), executes reference solutions across all four benchmark labs, and generates an automated Faculty Node Certificate (`FAC-NODE-PASS`).

4 Industrial Accreditation & Student Credential Lifecycle

Upon completing the four benchmark labs specified in ARC-SPEC-2026-EDU-02, a student's final pull request produces a cryptographically signed credential that bridges academic achievement directly with industrial workforce verification.

```
{
  "@context": ["https://www.w3.org/2018/credentials/v1"],
  "id": "urn:uuid:dae-2026-cert-9874-41a0",
```

```
  "type": ["VerifiableCredential", "CertifiedAdmissibilityEngineer"],
  "issuer": "did:arcstone:path-a-verifier-node-01",
  "credentialSubject": {
    "id": "did:github:student-username",
    "accreditation": "Certified Admissibility Engineer (CAE-Level 1)",
    "verificationMetrics": {
      "spec_compliance": ["EDU-01", "EDU-02", "EDU-03"],
      "max_verified_wcet_ms": 7.89,
      "heap_allocation_bytes": 0
    }
  }
}
```

Listing 2: Certified Admissibility Engineer (CAE) Verifiable Credential

4.1 Three-Tiered Industrial Certification Ladder

1. **CAE-Associate (Level 1):** Achieved via academic coursework (Modules I & II). Demonstrates mastery of eBPF Ring-0 filtering, zero-heap constraints, and 5D state vector tracking.
2. **CAE-Professional (Level 2):** Achieved via full lab suite completion (Modules I–IV). Verifies ability to maintain sub-12ms hardware interlocks and Kingman-clamped queue dynamics under stress.
3. **CAE-Architect (Level 3):** Industry-level designation awarded when an engineer deploys an open-source Path A extension or edge node that achieves $C_{\text{ops}} = 0$ in live production environments.

SPECIFICATION COMPLETE: ARC-SPEC-2026-EDU-03
DEPLOYMENT POSTURE: ARCHIVED FOR FUTURE PATH A EXTENSION
AUTHORIZATION HASH: A-77-DELTA-SHIELD-LOCKED